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Breaking the mold: how visual atypicality shapes brand-generated image virality

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ABSTRACT

Brands frequently post images on social media, hoping consumers will share them to enhance product visibility and brand awareness. This research introduces *visual atypicality* (i.e., the extent to which an image's visual presentation of a brand's products deviates from prevailing industry norms) and examines its impact on the virality of brand-generated images. Employing machine learning techniques to analyze 60,669 images from 330 apparel brands and 4,087 images from 27 car brands, supplemented by an online experiment, this research reveals that the effectiveness of visual atypicality depends on brand heritage. For low-heritage brands, visual atypicality enhances virality by increasing entertainment value. Conversely, for high-heritage brands, visual atypicality reduces virality as it clashes with consumer expectations. These findings suggest that, in order to maximize social media engagement, brands should strategically consider the degree to which their content aligns with both industry conventions and their own brand identity.

1. Introduction

Social media has emerged as a crucial communication channel for brands seeking to engage consumers (Hudson et al., 2016; Zhang et al., 2017). Advances in image creation and editing technologies, coupled with faster internet speeds, have enabled brands to leverage vivid, high-quality imagery to engage consumers online (Kanuri et al., 2024; Y. Li & Xie, 2020; Overgoor et al., 2022). In particular, brands strive to encourage consumers to share their images across social media platforms, driving image virality (Heimbach & Hinz, 2016), and, in turn, enhancing product visibility (Nguyen & Chaudhuri, 2019) and brand awareness (Colicev et al., 2019).

Notably, brands within a given industry often adopt a standardized and widely used visual presentation approach for showcasing their products in images. This approach is typically cost-effective, easy to produce, and ensures that images remain both aesthetically appealing and informative. For example, in the apparel industry, it is common to see models wearing a brand's clothing while standing centrally against a plain background, a presentation approach that not only ensures clarity in product display but also meets the basic aesthetic expectations of online consumers. Such images, which conform to prevailing industry standards, are considered visually typical. In contrast, brands may choose to deviate from these norms by incorporating distinctive visual elements (e.g., dynamic backgrounds, unconventional model poses, or unique camera angles) to create more visually atypical images.

However, the impact of such visual deviations from industry norms in brand-generated images on consumer engagement remains unclear. On one hand, research in advertising indicates that unconventional content is often perceived as more entertaining (Smith

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et al., 2008) and interesting (Schnurr, 2017), leading to more positive consumer attitudes and higher engagement (e.g., increased ad-viewing intention and greater willingness to buy; Yang & Smith, 2009). On the other hand, studies in product design suggest that atypical stimuli may hinder processing fluency by conflicting with consumer expectations (Landwehr et al., 2011). Likewise, research on consumer reviews finds that atypical image–text pairings tend to diminish perceived review helpfulness (Ceylan et al., 2023). To reconcile these contrasting perspectives, the current research examines whether, when, and how visual atypicality in brand-generated images influences their virality on social media.

We propose that visual atypicality makes images more likely to be shared when used by low-heritage brands because it adds entertainment value. However, for high-heritage brands, visual atypicality makes images less likely to be shared because it disrupts the consistent visual style consumers expect from established brands. To test these hypotheses, we analyzed two large-scale datasets, one from the apparel industry and another from the automobile industry, and complemented these analyses with a controlled online experiment.

This research makes several important contributions to the literature on visual marketing strategy and content virality. First, it introduces and develops the construct of *visual atypicality*, and conceptually distinguishes it from related constructs (e.g., *visual novelty*, *visual creativity*). It further establishes and validates a theoretical framework showing how visual atypicality significantly influences consumers' propensity to share brand-generated content on social media. Second, this work moves beyond prior studies that have primarily examined the effects of isolated image attributes, such as color complexity (Kanuri et al., 2024) and visual complexity (Overgoor et al., 2022), by emphasizing the importance of norm-based comparison. It shows that the effectiveness of visual content depends not only on its intrinsic features but also on the extent to which it conforms to or deviates from shared visual conventions within an industry. This perspective offers a more ecologically valid and context-sensitive understanding of how consumers respond to brand-generated visuals in competitive digital environments. Third, the findings challenge the prevailing assumption that content strategies have uniform effects across brands. Instead, they uncover a crossover pattern: visual atypicality increases virality for low-heritage brands but reduces it for high-heritage brands. This pattern is explained through a dual-path mechanism in which consumer responses are shaped by both the entertainment value of the content and the content's alignment with brand-related expectations. This insight extends theory by showing that the effectiveness of brand-generated content depends on both content-level appeal and brand-level fit.

Methodologically, this research introduces a novel machine-learning-based approach to quantifying visual atypicality. By leveraging convolutional neural network embeddings of product-focused images and cosine distance metrics within thematically clustered image sets, this method enables scalable measurement of how brand visuals diverge from prevailing industry norms. This technique lays the groundwork for future studies to examine visual atypicality across diverse formats (e.g., video, virtual reality) and industries.

From a managerial perspective, the findings offer clear and actionable guidance for brands seeking to enhance image virality on social media. Rather than adopting a one-size-fits-all visual strategy, brands should consider their level of heritage when deciding how much to visually differentiate from industry norms. Low-heritage brands may benefit from breaking visual conventions to increase entertainment value and stand out, whereas high-heritage brands are better served by preserving visual conventions to align with consumer expectations. The following sections review the relevant literature, outline the theoretical framework, and present the empirical studies (Studies 1–3) that support these contributions.

Table 1
Overview of research on content virality drivers (ranked by year).

Study	Brand-generated content	Content format	Content features	Content–brand fit
Berger and Milkman (2012)		Text	Emotional intensity and arousal	
Stieglitz and Dang-Xuan (2013)		Text	Emotional valence in text	
Koch and Benlian (2015)	Yes	Text	Perceived scarcity and personalization	
Akpinar and Berger (2017)	Yes	Text	Emotional appeals	
Cruz et al. (2017)	Yes	Text	Use of second person pronouns	
Zhang et al. (2017)	Yes	Text	Content–user fit	
Lee, Hosanagar, and Nair (2018)	Yes	Text	Humorous and emotional content	
Ordenes et al. (2019)	Yes	Text, image	Interaction of image acts and message intentions	
Pancer et al. (2019)	Yes	Text	Readability	
Tellis et al. (2019)	Yes	Video	Elements in videos	
Li and Xie (2020)		Image	Presence of images	
Reich and Pittman (2020)	Yes	Text, image	Content–platform fit	
McShane et al. (2021)	Yes	Text	Use of emojis	
Pezzuti, Leonhardt, and Warren (2021)	Yes	Text	Certainty in language	
Abell and Biswas (2023)		Image	Presentation of healthy foods and persons in images	
Kanuri et al. (2024)		Image	Color complexity of images	
The current research	Yes	Image	Visual atypicality of images	Yes

2. Theoretical background

2.1. Content virality on social media

Extensive research has explored the factors that drive content virality on social media (see Table 1 for a comprehensive summary). Much of this research stream has focused on textual content, examining features such as the expression of certainty (Pezzuti et al., 2021), the use of second-person pronouns (Cruz et al., 2017), and the use of emojis (McShane et al., 2021). Advancements in image-generation technology and faster internet speeds have fueled the proliferation of visual content online, leading to an increased scholarly focus on image-specific factors that drive consumer engagement. For example, Li and Xie (2020) showed that simply adding images to a social media post significantly increases the shareability of the post. Abell and Biswas (2023) found that images featuring healthy food alongside individuals could enhance consumer engagement, including sharing behaviors. Kanuri et al. (2024) revealed that greater color complexity in images tends to capture attention and stimulate higher levels of user engagement, including content sharing.

However, thus far, existing research on image virality has predominantly examined how specific visual attributes (e.g., imagery's color or visual complexity; Kanuri et al., 2024; Overgoor et al., 2022) influence consumer responses. Little attention has been paid to how an image's alignment with industry norms, or its deviation from them, affects virality. Furthermore, prior studies have largely overlooked the interplay between content and the brand's profile, and how this alignment, or lack thereof, impacts content virality. This research seeks to address these critical gaps.

2.2. Typicality vs. atypicality

To understand *atypicality*, it is essential first to define its opposite, *typicality*. *Typicality*, sometimes referred to as *prototypicality*, denotes the extent to which a stimulus represents a given category (Landwehr et al., 2013). Such a category applies to various domains, including product design (e.g., a coat typically has two sleeves rather than one), societal impressions of certain professions (e.g., a banker is typically envisioned wearing a suit rather than a pyjamas), and even abstract concepts (e.g., romance is typically symbolized by roses rather than lotuses). *Typicality* reflects the central representation of a category, embodying the features that people expect (Langlois & Roggman, 1990; Minda & Smith, 2011; Rosch & Mervis, 1975). Conversely, *atypicality* refers to deviations from these established norms, encompassing features that are less common or unexpected within a given category.

Marketing research has examined atypicality primarily in product design, examining how deviations from conventional aesthetics influence consumer preferences. For example, research has assessed how much a brand's car design diverges from the average and typical design and its effect on consumer perceptions (Landwehr et al., 2011). Other studies have investigated atypicality in the context of advertising strategies, idea generation, and consumer spending patterns. See Table W1 in Web Appendix A for a summary of relevant literature.

2.3. Visual atypicality in brand-generated images

Brands increasingly rely on images to present their products across both print and digital media platforms (Klostermann et al., 2018; Overgoor et al., 2022). We propose that, over time, brands within a particular industry spontaneously converge on a typical way of visually presenting their products. This proposition is grounded in *organizational learning theory*, which posits that organizations learn from experience, gradually refining their practices through experimentation and adaptation to the environment to enhance effectiveness (Argyris, 1977; Dodgson, 1993; Levitt & March 1988). Through iterative testing and consumer feedback, brands identify the most effective visual presentation strategies that satisfy both aesthetic and informational needs of target audiences. These strategies, once proven effective, diffuse across the industry and solidify into widely accepted visual norms.

For example, automotive brands may experiment with different photographic techniques by varying framing, lighting, and background composition. Over time, they often find that positioning a car centrally against a clean, minimal background improves clarity and meets basic consumer expectations. Because this approach is cost-effective and consistently appealing, it becomes widely adopted and codified as an industry standard. Conversely, deviations from these established norms, such as placing a car at the edge of the frame against a complex background, are perceived as unconventional and categorized as *atypical*.

Such patterns of standardization are not unique to brand imagery. Similar processes occur in other visual domains, where design norms emerge through collective learning and user adaptation. For example, website interfaces have evolved through extensive user experience research and iterative testing. Designers have tried various layouts, navigation structures, and visual elements to optimize usability and engagement. Over time, conventions such as top navigation bars, clear content hierarchies, and familiar color schemes have become standard because they reliably meet user expectations and enable intuitive browsing (Kalbach, 2007; Tuch et al., 2012).

Based on this understanding, we define *visual atypicality in brand-generated images* as the extent to which an image's visual presentation of a brand's products deviates from established industry norms within a specific thematic and temporal context. This definition recognizes that visual atypicality must be evaluated relative to three boundary conditions: industry context, image theme, and temporal context. First, *industry context* is critical, as different industries serve distinct audiences and product categories, each with its own expectations for visual communication. A visual approach that appears normal in one industry may be entirely disruptive in another. Second, *image theme* matters because visual atypicality concerns how products are visually presented ("how they are presented differently"), not the atypicality of the products themselves ("what is presented differently"). To assess this accurately, the focal product type must be held constant across the images being compared. For example, to determine whether a T-shirt is presented in a

visually atypical way, the image should be compared with other images that also feature T-shirts. Third, *temporal context* is important because visual norms are not fixed; they evolve in response to technological, social, and cultural shifts (e.g., the rise of minimalistic visual styles as a cultural reaction to material excess and overconsumption; Wilson & Bellezza, 2022). While such shifts shape norms over time, they usually unfold gradually. In other words, visual norms tend to remain relatively stable over shorter periods, such as within a single calendar year. Accordingly, to capture meaningful deviations, visual atypicality should be assessed within a clearly defined and stable temporal window.

To further clarify the construct, it is necessary to distinguish *visual atypicality* from related concepts such as *visual novelty* and *visual creativity*. *Visual novelty* refers to the degree to which a visual stimulus is perceived as new (i.e., something not previously seen or encountered; Kim & Lakshmanan, 2015; Sung et al., 2016). This inherently involves temporal comparison and is typically judged against an individual's prior exposure. For example, *Impressionism* was novel when it first emerged because it departed from prevailing academic painting techniques. In contrast, *visual atypicality* does not necessarily require such temporal newness. A visual presentation approach that was once common (e.g., black-and-white imagery) may no longer be novel today, but can still be considered atypical if it diverges from current norms in an industry where most visuals are now colorful (e.g., the contemporary fashion sector, where most visuals are colorful rather than black-and-white).

Visual creativity refers to the extent to which a visual presentation is both original and meaningful (Jiang et al., 2024; Smith & Yang, 2004). Unlike visual atypicality, creativity is not necessarily evaluated within an industry-specific context. A particular visual style may be seen as broadly creative yet remain typical within an industry. For example, surrealist, abstract, or mysterious visuals are often considered creative in general but may be standard in luxury fashion advertising (Dion & Arnould, 2011; Kapferer, 2014). Conversely, visuals that are not widely regarded as creative may still be atypical within a given industry. For instance, minimalist, monochromatic visuals may not be inherently creative but can be atypical in sectors that usually favor bold, vibrant imagery, such as children's toy marketing. Moreover, an essential criterion for creativity is perceived meaningfulness (i.e., whether the departure from convention is seen as meaningful and valuable by the audience; Hofstetter et al., 2021; Rosengren et al., 2020). Such judgments are subjective and often vary across individuals. In contrast, visual atypicality depends solely on the degree of deviation from prevailing industry norms, regardless of whether that deviation is perceived as meaningful or valuable.

2.4. Brand heritage

Prior research suggests that content strategies are not universally effective; rather, their success often depends on contextual and brand-specific factors (Montaguti et al., 2023). In line with this perspective, it is crucial to consider brand-level characteristics when examining how visual atypicality in brand-generated images affects consumer responses. To this end, we introduce *brand heritage* as a key moderating factor in our theoretical framework. Initially conceptualized by Aaker (1997) as a core component of brand identity, the concept of brand heritage has been further developed in marketing literature.

Consistent with prior research, we define brand heritage as a temporal and symbolic dimension of brand identity that reflects a brand's longevity, historical continuity, core values, and the perceived significance of its past (Pecot et al., 2019; Urde et al., 2007). While chronological age (or brand longevity) serves as a readily observable and widely accepted proxy for brand heritage (Pecot et al., 2019; Urde et al., 2007), it is only one facet of a broader construct. Brand heritage also encompasses the stability of core brand values over time, the ability to adapt without compromising brand identity, and the extent to which a brand's past is perceived as meaningful in shaping its current and future identity.

Extensive evidence suggests that consumers form intuitive beliefs and associations linking brand heritage with positive attributes. Brands with a strong heritage are often perceived as more credible, trustworthy, and higher in quality (Balmer & Burghausen, 2015; Pecot et al., 2019; Urde et al., 2007; Wiedmann et al., 2011). As such, brand heritage can serve as a valuable source of competitive advantage. When effectively communicated, it can enhance brand loyalty, increase purchase intentions, and even justify price premiums (Rose et al., 2016; Wiedmann et al., 2011). Many firms strategically emphasize their heritage in branding and marketing communications. For example, French's, the American mustard brand, prominently displays "Since 1904" in its logo to highlight its long history and reinforce consumers' perceptions of reliability and quality.

With the increasing prominence of social media as a marketing platform, both high-heritage and low-heritage brands are creating digital content to engage consumers (Butcher & Pecot, 2021). However, existing research provides limited guidance on how brands should adapt their content strategies to their level of heritage. It remains unclear whether brands with more or less heritage benefit equally from visually atypical content. This question is especially relevant in today's attention-driven media landscape, where consumers encounter diverse and dynamic brand imagery. This research addresses this issue by examining how visual atypicality in brand-generated images affects virality depending on brand heritage.

3. Hypothesis development

When deciding whether to share brand-generated content on social media, consumers typically make two distinct but related judgments. First, at the content level, they evaluate the content's intrinsic appeal, such as how entertaining it is (e.g., de Vries et al., 2012). Second, at the contextual level, they assess how well the content fits with their expectations of the brand, considering not only the content's features but also the brand's identity, positioning, and symbolic meaning (Montaguti et al., 2023). Building on this dual-layered framework, we predict that the effect of visual atypicality on image virality will differ based on the strength of the brand's heritage.

3.1. Visual atypicality increases the virality of low-heritage brands' images through enhanced entertainment

Prior research suggests that content that diverges from conventional norms tends to offer greater entertainment value (i.e., the extent to which content elicits amusement, interest, or enjoyment; Smith et al., 2008). Visually atypical brand-generated images, by departing from familiar visual patterns, are therefore likely to be perceived as more entertaining than visually typical ones. This heightened entertainment value of visually atypical images, in turn, is likely to drive consumers' content-sharing behavior on social media due to two key psychological reasons. First, the tendency to share entertaining content can be understood through the lens of *impression management*, which suggests that individuals are motivated to present themselves in ways that foster favorable impressions and minimize the risk of negative evaluation in everyday social interactions (Goffman, 1959). As social beings, individuals inherently want to be seen as engaging and socially attuned rather than boring or indifferent (Berger, 2014; Chen, 2017). This impression management motivation can extend to the digital sphere. Because the content shared on social media is visible to one's online network, it influences how the sharer is perceived by others (Buechel & Berger, 2018). As a result, entertaining (vs. boring) content is more likely to be shared, as sharing such content provides a low-effort yet effective means of projecting a positive and socially appealing persona to others (Buechel & Berger, 2018). Second, from a *social bonding* perspective, humans possess a fundamental need to form and maintain meaningful social connections (Baumeister & Leary, 1995). One way people fulfill this need is by sharing emotionally evocative content that creates shared emotional experiences and strengthens interpersonal bonds (Berger, 2014). Entertainment, by nature, is a positive emotional experience (Weidman & Tracy, 2020). Therefore, content that entertains should be more likely to be shared.

Empirical research supports the pivotal role of entertainment in driving content virality. Entertaining product information generates more spontaneous word-of-mouth than mundane information (Berger & Iyengar, 2013; Berger & Schwartz, 2011). Similarly, *New York Times* articles with greater entertainment value are shared more frequently among readers (Berger & Milkman, 2012), and entertaining video advertisements are more likely to go viral online (Tellis et al., 2019).

To move beyond a purely content-level analysis, we consider the brand-level context in which the content is embedded. Brands with low heritage (e.g., newly emerging startups) are often in the early, exploratory stages of their brand life cycle and typically lack historical depth. At this stage, the brand itself carries little symbolic meaning and is unlikely to serve as a strong cue shaping consumer responses (Balmer & Burghausen, 2015). As a result, consumers are likely to pay less attention to brand-level signals and focus more on the intrinsic qualities of the content. In other words, when brand heritage is low, the decision to share content depends less on how well it aligns with the brand's identity and more on the content's inherent appeal (e.g., its entertainment value). Taken together, we hypothesize:

H_{1a}: Visual atypicality enhances the virality of low-heritage brands' images.

H_{1b}: The positive effect of visual atypicality on the virality of low-heritage brands' images is mediated by increased entertainment value.

3.2. Visual atypicality decreases the virality of high-heritage brands' images due to incongruence with consumer expectations

When brand heritage is high, the evaluation process changes notably. High-heritage brands inherently carry substantial symbolic value (e.g., consistency, reliability, and enduring quality) that distinguishes them from low-heritage brands, which may be perceived as little more than names without deeper meaning (Urde et al., 2007; Wiedmann et al., 2011). Consumers who are aware of a brand's long-standing legacy are unlikely to disregard its symbolic influence when forming judgments (Rose et al., 2016; Wiedmann et al., 2011). This logic extends to social media contexts. For high-heritage brands, the symbolic meanings associated with heritage become especially salient. As a result, consumers evaluate brand-generated content not in isolation but in light of brand-level cues, particularly whether the content aligns with the brand's established identity.

Moreover, brand heritage provides a competitive advantage, as it often signals stability, reliability, and trustworthiness (Pecot et al., 2019; Urde et al., 2007; Wiedmann et al., 2011). Brands with strong heritage frequently emphasize these attributes in their marketing and branding to attract and reassure consumers (Balmer & Burghausen, 2015; Dion & Borraz, 2015). For instance, Patek Philippe, a watch brand founded in 1839, consistently uses its marketing to highlight traditional watchmaking, timeless craftsmanship, and a legacy passed down through generations. According to *schema theory*, repeated exposure to such branding shapes consumer expectations, reinforcing associations with stability, tradition, and continuity (Stayman et al., 1992). These expectations extend to all brand communications, including social media content, where consumers anticipate that high-heritage brands will share visuals that reflect these enduring qualities.

However, when a high-heritage brand adopts visually atypical imagery (e.g., by incorporating bold, unconventional, or experimental visual elements), this departure from established norms is likely to be seen as misaligned with the brand's expected identity. Such misalignment can, in turn, reduce content sharing for three main reasons. First, *schema incongruity theory* suggests that when content deviates from expectations, it requires greater cognitive effort to process, which can reduce emotional engagement (Fiske, 1982). When consumers encounter a visually atypical image from a high-heritage brand known for its conventional and consistent aesthetic, they are likely to experience cognitive dissonance (Lynch & Schuler, 1994). This dissonance may further evoke negative feelings, making them less likely to share the content. Second, *brand authenticity theory* emphasizes the importance of consistency between a brand's identity and its communications (Morhart et al., 2015). When high-heritage brands present content that deviates from established expectations, consumers may perceive this shift as a violation of brand authenticity (Cinelli & LeBoeuf, 2020). Prior research suggests that authenticity is a key driver of consumer engagement with brand-generated content (Eigenraam et al., 2021; Montaguti et al., 2023). If a high-heritage brand is seen as inauthentic, consumers may be less willing to engage with or share its content. Third, from an *identity-signaling* perspective (Berger & Heath, 2007; Packard & Wooten, 2013), consumers may share content

from high-heritage brands as a means of expressing their affinity for tradition and nostalgia, values that align with the brand's long-standing historical positioning. If the brand adopts visually atypical imagery that contradicts these expectations, the content loses this self-expressive function. As a result, consumers may be less willing to share visuals that no longer signal their identity.

Taken together, we argue that when brand heritage is high, consumers do not evaluate the content in isolation; instead, they pay greater attention to broader brand-related cues. In this context, visually atypical (vs. typical) images, despite their inherent entertainment value, are highly likely to violate expectations associated with high-heritage brands, making them less likely to go viral. We thus hypothesize:

H_{2a}: Visual atypicality decreases the virality of high-heritage brands' images.

H_{2b}: The negative effect of visual atypicality on the virality of high-heritage brands' images is mediated by reduced alignment with consumer expectations.

4. Overview of studies

Our hypotheses are tested by three empirical studies. Study 1 employs econometric modeling to analyze a dataset from the apparel

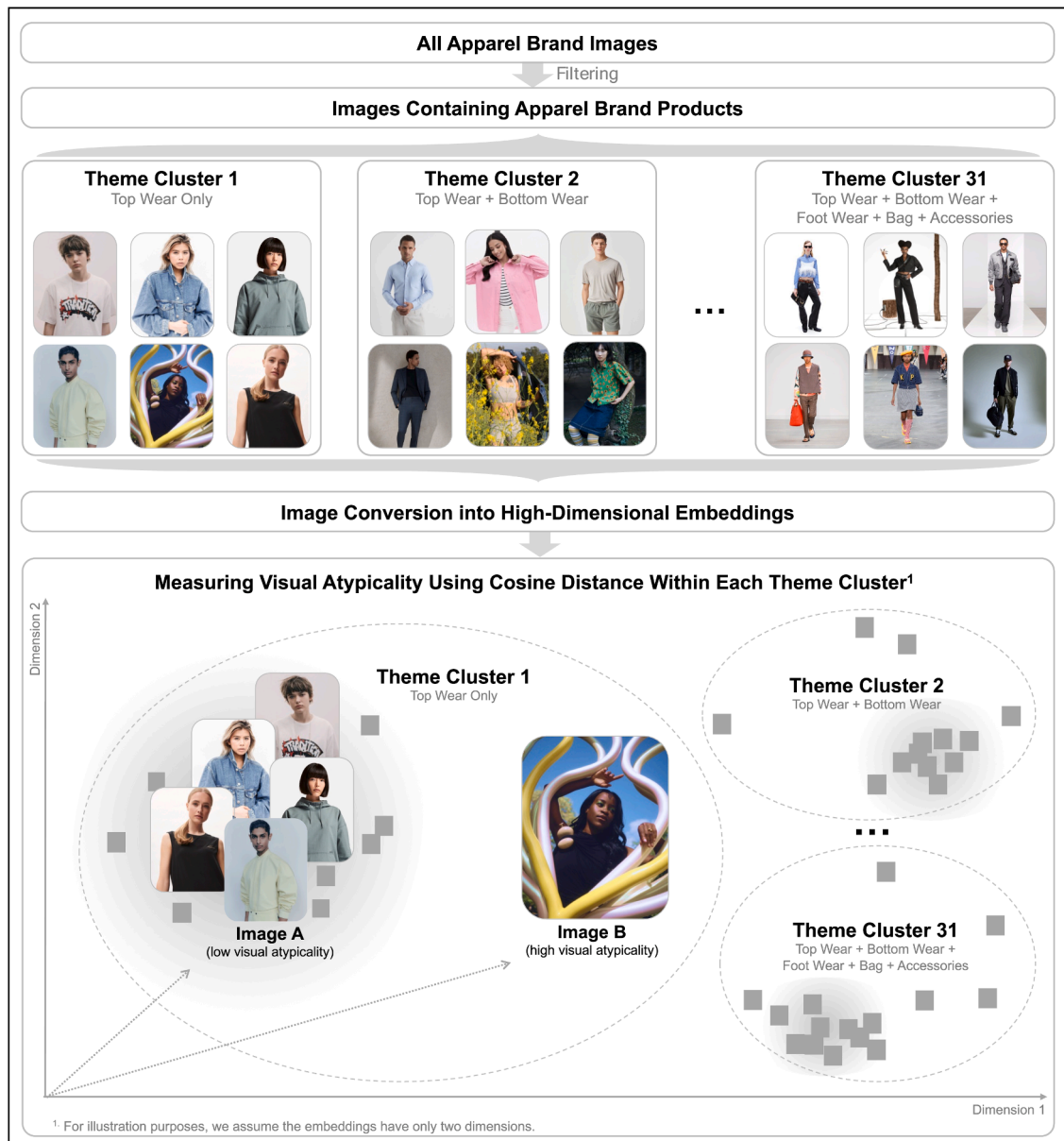


Fig. 1. Illustration of visual atypicality measurement process.

industry, providing initial empirical evidence for the main contrasting effects posed in our framework. To assess the generalizability of these effects across industries, Study 2 replicates the analysis using data from the automobile sector. Finally, Study 3 employs an online experiment to establish causality and further examine the underlying mechanism driving the observed effects.

5. Study 1: field evidence from the apparel industry

5.1. Data sources

This study aims to investigate the impact of visual atypicality in brand-generated images on their virality. We selected the apparel industry as our focal context for two primary reasons. First, apparel products are highly personal and relevant to consumers, making them more likely to generate significant social media engagement, which provides a rich dataset for analysis. Second, the apparel industry encompasses a diverse range of brands, from emerging ones (e.g., Jordanluca, founded in 2018) to well-established heritage brands (e.g., Levi's, founded in 1853), allowing us to examine variations in brand heritage.

To construct our dataset, we identified apparel brands using two influential online retail platforms: Zalando's UK (<https://www.zalando.co.uk>), known for its extensive collection of affordable fashion, and Farfetch's UK (<https://www.farfetch.com/uk>), a platform renowned for premium apparel. This dual-source approach ensured broad coverage of brands across various market segments. By scraping brand data from both platforms, we compiled a list of 1,145 apparel brands. Despite the comprehensiveness of these platforms, some iconic brands (e.g., Chanel, Zara) were missing, as their products were not available on either site. To address this limitation, we conducted an online survey via Amazon Mechanical Turk (MTurk), where 54 participants (77.78 % female, 22.22 % male; $M_{\text{age}} = 29.40$, $SD = 8.68$) were asked to list ten apparel brands they were familiar with. This process led to the identification of 30 additional brands, expanding our dataset.

5.2. Data collection from social media

To examine content virality, we sourced social media data from Twitter (now X), a widely used platform where brands frequently interact with consumers (Lu et al., 2022). We identified the official Twitter accounts for the brands in our dataset and used the Twitter API (<https://developer.x.com/en/docs/twitter-api>) to scrape the most recent 3,200 tweets (the upper limit set by the API) from each account, excluding replies and retweets while retaining tweets featuring images. All associated images were downloaded. Data collection was conducted in January 2023. Given that Twitter's API retrieves the most recent posts, the dataset predominantly reflects tweets from 2022. To maintain consistency, we restricted our dataset to images posted during that year.

Since some tweets contained multiple images, and our focus was on individual images while assuming that images within the same tweet generally share similar visual presentation approaches, we curated the dataset in a "one-image-per-record" structure. This approach ensured that each record corresponded to a unique image, along with its associated metadata, including the number of shares (retweets). At this stage, the dataset comprised 72,129 images from 330 apparel brands. To measure visual atypicality more precisely, we further refined the dataset in the next step.

5.3. Refining the dataset and measuring visual atypicality (explanatory variable)

The explanatory variable in this study is the degree of visual atypicality in brand-generated images. Prior research quantifies atypicality (e.g., product design atypicality) by measuring deviations from category norms within the same industry. For instance, Landwehr et al. (2011) assessed design atypicality by calculating the Euclidean distance between a car's design and a prototypical car image, with larger distances indicating greater atypicality. Similarly, Liu et al. (2017) assessed product design atypicality by determining how far a product deviated, in Euclidean distance, from the central design of its segment (e.g., economy or luxury). Yang, Zhang, and Kannan (2022) applied cosine distance to identify brands that significantly deviated from others in engagement patterns. Inspired by these studies, we employed a similar approach to quantify visual atypicality (see Fig. 1 for a detailed overview of the measurement process).

As mentioned in the Theoretical Background section, visual atypicality is distinct from content atypicality. To accurately measure visual atypicality, we must ensure that the content remains consistent across the images being compared. Hence, the first step in our approach involved clustering images based on their primary content (i.e., products), which were detected by the Google Vision API's "Detect Multiple Objects" function (available at: <https://cloud.google.com/vision/docs/object-localizer>). The detected objects were classified into six categories: five apparel-related categories ("top wear," "bottom wear," "footwear," "bag," and "accessories") and a miscellaneous category ("others"). Since the "others" category was too diverse to form meaningful clusters, we excluded these images from further analysis. Approximately 84.11 % of images contained identifiable products (i.e., at least one object classified under "top wear," "bottom wear," "footwear," "bag," or "accessories"), resulting in a final dataset of 60,669 images from 330 apparel brands.

Next, we cataloged all unique combinations of the five main product categories (e.g., top wear alone, top wear + bottom wear, through to all five categories together). This approach yielded $31 (2^5 - 1 = 31)$ distinct theme-based image clusters. See Table W2 in Web Appendix B for the distribution of sample images by theme. After clustering, images were transformed into high-dimensional embeddings using the VGG-16 neural network model (Simonyan & Zisserman, 2015), pre-trained on ImageNet. These embeddings capture the visual characteristics of each image and allow for comparison in a high-dimensional space.

Visual atypicality was then computed as the average cosine distance of each image from all other images in the same theme. For a given image i , its visual atypicality is calculated as:

$$VisualAtypicality_i = \frac{1}{N_k - 1} \cdot \sum_{j=1, j \neq i}^{N_k - 1} \left(1 - \frac{E(Image_i) \cdot E(Image_j)}{|E(Image_i)| |E(Image_j)|} \right) \quad (1)$$

where N_k represents the total number of images in the theme k , and $E(Image_i), E(Image_j)$ are embedding vectors for images i and j , respectively.

A smaller average cosine distance indicates that an image closely aligns with the cluster's representative and typical visual presentation, while a larger distance suggests a greater deviation from the majority in terms of its visual presentation. For example, in Fig. 1, Image A depicts a model wearing top wear, centered in the frame against a plain background, which is a conventional presentation among apparel brands, resulting in a small average cosine distance, making Image A typical and representative of its theme. In contrast, Image B, captured from an unconventional camera angle with a distinctive model pose and a stylish decorative pattern, exhibits a larger average cosine distance, making it relatively atypical within the cluster. Additional examples of visually typical and atypical apparel brand images are presented in Table W3 of the Web Appendix B.

To ensure the reliability of this automatic algorithm-based visual atypicality measure, we conducted two validation studies. The first compared our automated scores with human judgments of image typicality, using a pairwise comparison task in which MTurk participants evaluated 600 image pairs sampled from three common product themes. The second assessed the temporal stability of industry-level visual norms by recalculating atypicality scores based on quarterly subsets (Q1–Q4) of the 2022 dataset and examining consistency across time. Detailed procedures, descriptive statistics, and results of the two validation studies are reported in Web Appendix B.

5.4. Measuring image virality (dependent variable)

The potential for a brand-generated image to achieve virality is largely contingent upon consumers' willingness to share it online. Following existing research, we used the number of shares as a proxy for content virality (Jalali & Papatla, 2019; Tellis et al., 2019). On social media platforms, brand-generated images are disseminated through posts (e.g., tweets), and the number of shares (e.g., retweets) a tweet receives serves as an indicator of the image's virality.

5.5. Measuring brand heritage (moderator) and defining high vs. low heritage

Brand heritage reflects a brand's historical value, primarily determined by its longevity in the market (Pecot et al., 2019). A brand with over a century of history, for example, should possess a greater perceived heritage than one founded more recently. We operationalized brand heritage by calculating a brand's age as the difference between the year of data collection (2023) and its founding year.

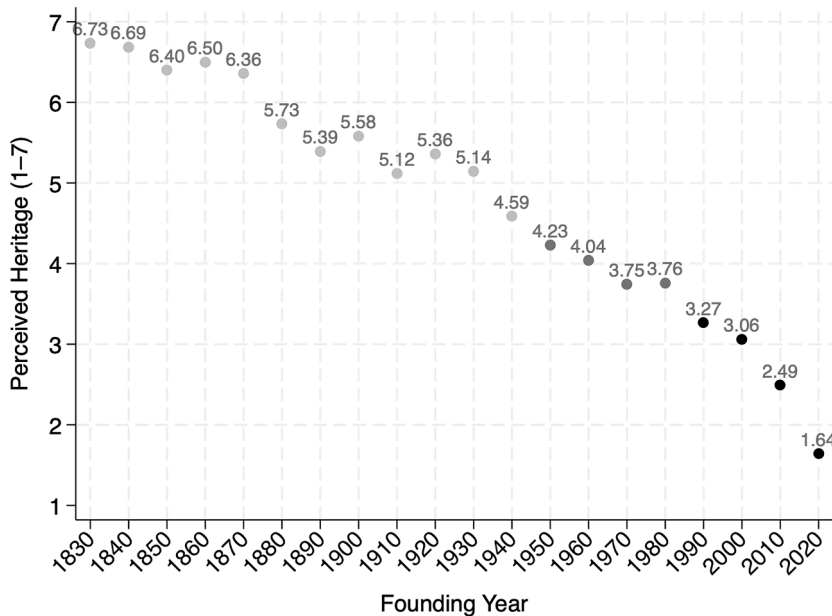


Fig. 2. Classification of high- and low-heritage apparel brands based on perceived heritage ratings. Notes: Dots show mean perceived heritage. Light gray dots indicate means significantly above 4 (high heritage), black dots indicate means significantly below 4 (low heritage), and dark gray dots indicate means not significantly different from 4 (mid-level heritage).

In our analyses, we aim to examine the effect of visual atypicality on image virality separately for low- and high-heritage brands. To establish an objective threshold for distinguishing between high- and low-heritage brands, we adopted a crowdsourcing approach using an online survey. The survey began with a brief introduction to the concept of brand heritage, followed by a prompt: “Imagine an apparel brand founded in [YEAR, e.g., 1830, 1840, 1850 ... 2020]. It has been operating continuously in the market. Based on this information alone, how much heritage do you think this apparel brand has? (1 = Low Heritage, 7 = High Heritage).” To ensure reliable estimates, we recruited 200 participants via Prolific (42.50 % female, 56.50 % male, and 1.00 % others; $M_{age} = 31.08$, $SD = 10.56$), with each participant evaluating five different founding years.

The survey results are displayed in Fig. 2, which reveal two key insights. First, perceived brand heritage gradually decreased as the founding year increased, validating our approach of using brand age as a proxy for brand heritage. Second, the founding years in our dataset covered a wide spectrum of perceived heritage, ranging from very high ($M_{year=1830} = 6.73$) to very low ($M_{year=2020} = 1.64$). This suggests that our dataset provides a comprehensive representation of apparel brands with varying levels of heritage.

More importantly, this survey allowed us to establish thresholds for distinguishing high- and low-heritage brands. Using a midpoint rating of 4 as a reference, we conducted *t*-tests on the mean perceived heritage ratings for each founding year. Brands with founding years that had a significantly higher mean perceived heritage rating than 4 were classified as high-heritage brands (highlighted by light gray dots in the graph), while those with significantly lower ratings were categorized as low-heritage brands (highlighted by black dots in the graph). Brands that fell within the middle range were highlighted in dark gray. Based on this classification, apparel brands established in or before 1940 were considered high-heritage brands, while those founded in or after 1990 were classified as low-heritage brands.

5.6. Control variables

To exclude potential confounding effects from other factors, we controlled for a comprehensive list of features (variable names in *italics*). First, we accounted for brand-specific characteristics (see Web Appendix C for the detailed methodology for measuring each brand-level variable). Prior research has shown that brand heritage can be an important source of a brand’s luxury perception (Dion & Borraz, 2015). Therefore, we controlled for brand luxury level (*BrandLuxury*). Besides, many contemporary apparel brands with long histories originate from Europe (e.g., Gucci, founded in 1921), so we also accounted for a brand’s European origin (*EuropeanBrand*).

We also measured and controlled for two core personality traits of apparel brands: femininity (*BrandFemininity*) and sportiveness (*BrandSportiveness*). Femininity refers to the extent to which a brand focuses on female consumer segments, with highly feminine brands primarily selling women’s wear (e.g., Chanel) and low-femininity brands mainly targeting men’s wear (e.g., Zegna). Sportiveness, on the other hand, indicates the extent to which a brand focuses on sportswear, with highly sportive brands specializing in athletic apparel (e.g., Nike) and low-sportiveness brands producing apparel primarily for non-sport occasions (e.g., Dolce & Gabbana). Since brand popularity (*BrandPopularity*) is a key driver of engagement levels, we also controlled for this factor by log-transforming the total number of Twitter followers each brand had at the time of data collection.

Next, we accounted for image-related features that could potentially confound the main effect (see Web Appendix D for the detailed methodology of image-level variables). Our core construct, visual atypicality, captures inter-brand visual atypicality. However, since such atypicality may either align with or deviate from a brand’s own visual norms, we also measured intra-brand visual atypicality (*IntraBrandAtypicality*) using the same methodology as inter-brand visual atypicality but considering only images from the same brand. Besides, as visual atypicality should not compromise an image’s aesthetic appeal (*ImageAesthetics*) or quality (*ImageQuality*), we employed a pre-trained machine-learning model to assess an image’s aesthetic level, and we measured absolute quality by the image’s overall pixel volume. We also captured the images’ aspect ratio (*ImageAspectRatio*) to account for the potential influence of formatting variations on image virality. We also accounted for the number of people appearing in an image (*NumberOfFaces*) and the presence of sensitive content (e.g., violent imagery; *Sensitive*), as both elements are highly relevant to engagement.

Furthermore, since visual atypicality does not necessarily equate to product atypicality, we controlled for product-related features (see Web Appendix E for measurement details). Specifically, we accounted for the number of products (*NumberOfProducts*) displayed in the image and the visual characteristics of these products, including color saturation (*ProductSaturation*), brightness (*ProductBrightness*), contrast of brightness (*ProductContrastOfBrightness*), color complexity (*ProductColorComplexity*), and color warmth levels (*ProductWarmth*).

Given that social media images are often accompanied by text, we also controlled for text-related features (see Web Appendix F for measurement details). These included the length of text (*WordCount*), the presence of hashtags (*Hashtags*), at-mentions (*AtMentions*), URLs (*URLs*), and emojis (*Emojis*), as well as the emotionality of the accompanying text (*TextEmotionality*).

Weekend posting was accounted for using a binary variable indicating whether an image was posted on the weekend. *Season Fixed Effects* captured seasonal variations in consumer preferences by including controls for spring, summer, autumn, and winter. *Theme Fixed Effects* were included to account for the stratified nature of the visual atypicality measurement method (i.e., visual typicality scores were calculated within thematic categories). *Format fixed effects* controlled for differences in post formats, specifically whether posts contained one to four images.

5.7. Descriptive statistics

Table 2 presents the descriptive statistics for all non-factor variables, including their mean, standard deviation, minimum, median, and maximum values. The statistics are first displayed for the overall dataset, encompassing brands of all heritage levels. For facilitating comparison, we also provide separate descriptive statistics for low-heritage brands (i.e., those established in or after 1990) and

Table 2
Descriptive statistics (apparel brands).

	All Brands (N = 60,669)					Low Heritage Brands (N = 24,634)					High Heritage Brands (N = 10,952)				
	Mean	SD	Min	Median	Max	Mean	SD	Min	Median	Max	Mean	SD	Min	Median	Max
Share	57.73	1,203.27	0.00	2.00	127,121.00	15.98	351.88	0.00	1.00	41,654.00	166.32	2,478.81	0.00	3.00	127,121.00
VisualAtypicality	0.75	0.05	0.55	0.75	0.89	0.74	0.05	0.55	0.74	0.88	0.75	0.05	0.57	0.75	0.88
BrandHeritage	51.51	38.25	2.00	4.00	193.00	2.30	8.74	2.00	19.00	33.00	119.39	27.12	84.00	113.00	193.00
<i>Brand-Level Controls</i>															
BrandLuxury	2.14	0.56	1.02	2.16	3.60	2.09	0.54	1.04	2.11	3.32	2.34	0.65	1.02	2.09	3.60
EuropeanBrand	0.64	0.48	0.00	1.00	1.00	0.50	0.50	0.00	0.00	1.00	0.80	0.40	0.00	1.00	1.00
BrandFemininity	0.16	0.02	0.07	0.16	0.22	0.15	0.02	0.07	0.15	0.22	0.16	0.02	0.10	0.16	0.21
BrandSportiveness	0.14	0.03	0.06	0.13	0.39	0.14	0.03	0.06	0.13	0.39	0.14	0.04	0.08	0.13	0.30
BrandPopularity(Log)	4.95	1.07	0.48	4.98	7.13	4.45	1.00	0.48	4.56	7.07	5.25	1.02	0.90	5.05	7.13
<i>Image-Level Controls</i>															
IntrabrandAtypicality	0.68	0.13	0.00	0.71	0.90	0.66	0.15	0.00	0.69	0.90	0.69	0.12	0.00	0.71	0.90
ImageAesthetics	5.25	0.24	3.84	5.25	6.23	5.23	0.24	4.14	5.23	6.18	5.25	0.23	4.20	5.25	6.23
ImageQuality(Log)	6.03	0.11	4.62	6.06	6.16	6.04	0.10	4.62	6.06	6.16	6.03	0.09	5.01	6.06	6.16
ImageAspectRatio	0.91	0.27	0.41	0.80	5.38	0.92	0.26	0.41	0.80	4.00	0.92	0.31	0.55	0.80	5.38
NumberOfFaces	1.07	1.31	0.00	1.00	10.00	0.93	1.19	0.00	1.00	10.00	1.16	1.44	0.00	1.00	10.00
Sensitive	0.14	0.35	0.00	0.00	1.00	0.13	0.34	0.00	0.00	1.00	0.10	0.30	0.00	0.00	1.00
<i>Product-Level Controls</i>															
NumberOfProducts	4.27	2.60	1.00	4.00	10.00	4.18	2.66	1.00	4.00	10.00	4.42	2.56	1.00	4.00	10.00
ProductSaturation	80.38	43.67	0.00	76.87	251.38	73.01	43.57	0.00	69.60	251.38	85.25	41.77	0.00	81.51	241.19
ProductBrightness	132.23	45.55	5.20	131.02	254.31	136.81	46.70	7.50	135.05	254.31	124.59	43.20	5.20	123.36	242.07
ProductContrastOfBrightness	59.02	16.46	2.54	58.97	120.60	59.92	17.97	2.54	59.79	120.60	58.82	14.77	8.07	58.88	119.76
ProductColorComplexity	8.04	1.51	1.39	8.30	12.39	7.79	1.71	1.53	8.15	12.12	8.17	1.23	2.23	8.31	11.79
ProductWarmth	0.48	0.28	0.00	0.45	1.00	0.46	0.29	0.00	0.42	1.00	0.47	0.28	0.00	0.45	1.00
<i>Text-Level Controls</i>															
WordCount	28.65	11.22	5.00	27.00	63.00	27.18	11.34	5.00	25.00	61.00	31.83	10.11	5.00	32.00	63.00
Hashtags	1.71	2.05	0.00	1.00	23.00	1.58	2.00	0.00	1.00	21.00	2.12	2.29	0.00	1.00	23.00
AtMentions	0.34	0.76	0.00	0.00	14.00	0.28	0.67	0.00	0.00	10.00	0.36	0.85	0.00	0.00	10.00
URLs	0.65	0.61	0.00	1.00	8.00	0.68	0.64	0.00	1.00	6.00	0.62	0.52	0.00	1.00	8.00
Emojis	0.38	0.77	0.00	0.00	14.00	0.42	0.78	0.00	0.00	7.00	0.31	0.77	0.00	0.00	6.00
TextEmotionality	9.63	4.39	1.72	9.09	44.44	9.91	4.50	1.72	9.38	44.44	8.75	4.05	1.85	8.00	38.46
<i>Temporal Control</i>															
Weekend	0.22	0.41	0.00	0.00	1.00	0.20	0.40	0.00	0.00	1.00	0.23	0.42	0.00	0.00	1.00

high-heritage brands (i.e., those established in or before 1940). In addition, the pairwise correlation table for all continuous variables is displayed in Table W8 in [Web Appendix G](#).

5.8. Model specifications

To investigate the influence of visual atypicality on image virality, it was essential to select an econometric model suited to the characteristics of the dataset. The dependent variable, image virality (measured by the number of shares), is a count variable consisting of non-negative integers. As shown in the descriptive statistics ([Table 2](#)), the distribution of shares exhibits significant overdispersion with a long-tail distribution. Given this, we employed the negative binomial regression model, a widely used approach in social media engagement research ([Hartmann et al., 2021](#); [Hughes et al., 2019](#); [Ordenes et al., 2019](#); [Overgoor et al., 2022](#); [Rietveld et al., 2020](#)). This model extends Poisson regression by relaxing the assumption that the mean and variance are equal, making it appropriate for overdispersed data ([Rietveld et al., 2020](#)).

Using this negative binomial regression, a maximum likelihood estimation approach, we systematically tested our hypotheses through the following model specifications. First, we analyzed the full dataset (*Model 1*), which includes brands across all heritage levels. In this model (see *Equation (2)*), sharing volume serves as the dependent variable, while visual atypicality is the primary explanatory variable. To examine the moderating effect of brand heritage, we introduced brand heritage and an interaction term between visual atypicality and brand heritage into the model. All relevant control variables were also included.

$$\ln(E[\text{Shares}_i]) = \beta_0 + \beta_1 \cdot \text{VisualAtypicality}_i + \beta_2 \cdot \text{BrandHeritage}_i + \beta_3 \cdot (\text{VisualAtypicality}_i \times \text{BrandHeritage}_i) + \gamma'X_i + \delta_{t(i)} + \theta_{s(i)} + \phi_{f(i)} + \epsilon_i \quad (2)$$

where $E[\text{Shares}_i]$ is the expected number of shares for image i . X_i is a vector of standardized control variables; $\delta_{t(i)}$, $\theta_{s(i)}$, $\phi_{f(i)}$ represent theme fixed effects, season fixed effects, and format fixed effects, respectively; ϵ_i is the error term.

After conducting this full-sample analysis, we ran separate regressions to examine the effects of visual atypicality on image virality for low-heritage brands (i.e., those established in or after 1990; *Model 2*) and high-heritage brands (i.e., those established in or before 1940; *Model 3*). In these subgroup analyses (see *Equation (3)*), sharing volume and visual atypicality remained the key dependent and explanatory variables, while all other control variables were retained. To facilitate interpretation and ensure comparability across variables, all continuous variables except for the dependent variable (i.e., sharing volume) were standardized (i.e., transformed into z-scores with a mean of 0 and a standard deviation of 1) for the regression analyses.

$$\ln(E[\text{Shares}_i]) = \beta_0 + \beta_1 \cdot \text{VisualAtypicality}_i + \gamma'X_i + \delta_{t(i)} + \theta_{s(i)} + \phi_{f(i)} + \epsilon_i \quad (3)$$

5.9. Results

The estimation results for the three regression models in our main analysis are shown in [Table 3](#) (first three columns). When considering tweets from all brands, we found that the interaction term between visual atypicality and brand heritage was statistically significant ($\beta_{\text{VisualAtypicality} \times \text{BrandHeritage}} = -0.09$, $SE = 0.01$, $p < 0.001$), indicating that the impact of visual atypicality on image virality is dependent on the level of brand heritage.

Further subgroup regression analyses for low-heritage and high-heritage brands provide more direct support for our hypotheses. First, when brand heritage is low, visual atypicality was significantly and positively associated with sharing volume ($\beta_{\text{VisualAtypicality}} = 0.22$, $SE = 0.02$, $p < 0.001$), suggesting that for low-heritage brands, increasing visual atypicality tends to boost an image's virality on social media, supporting **H_{1a}**. In contrast, when brand heritage is high, the relationship between visual atypicality and sharing volume was significantly negative ($\beta_{\text{VisualAtypicality}} = -0.09$, $SE = 0.02$, $p < 0.001$), indicating that for high-heritage brands, increased visual atypicality tends to reduce image virality, supporting **H_{1b}**.

5.10. Robustness check with single-image tweets

In our main analysis, we included all types of tweets containing one to four images, using a tweet's total share count as a proxy for the virality of the focal image. Although we controlled for format with fixed effects, the share count could still be influenced by other images in the same tweet. To address this, we ran a robustness check using only single-image tweets, reducing the dataset to 23,418 images and ensuring that the share count directly reflects each image's virality.

We applied the same regression models to this refined dataset. Because all tweets contained only one image, format fixed effects were unnecessary and not included. The results ([Table 3](#)) confirmed the robustness of our main findings: the interaction between visual atypicality and brand heritage remained negative and significant ($\beta_{\text{VisualAtypicality} \times \text{BrandHeritage}} = -0.11$, $SE = 0.01$, $p < 0.001$). Consistent with **H_{1a}**, visual atypicality was positively associated with virality for low-heritage brands ($\beta_{\text{VisualAtypicality}} = 0.17$, $SE = 0.03$, $p < 0.001$). In contrast, supporting **H_{1b}**, visual atypicality was negatively related to virality for high-heritage brands ($\beta_{\text{VisualAtypicality}} = -0.17$, $SE = 0.05$, $p < 0.001$). These findings further reinforce the robustness and consistency of our main results.

5.11. Discussion

Leveraging a large-scale dataset from apparel brands, this study suggests that the impact of visual atypicality on image virality is

Table 3

Model estimation results for apparel brands.

	Main analysis (all tweets)			Robustness check (single-image tweets)		
	Model 1 (All Brands)	Model 2 (Low Heritage Brands)	Model 3 (High Heritage Brands)	Model 1 (All Brands)	Model 2 (Low Heritage Brands)	Model 3 (High Heritage Brands)
VisualAtypicality	0.12*** (0.01)	0.22*** (0.02)	-0.09*** (0.02)	0.11*** (0.02)	0.17*** (0.03)	-0.17*** (0.05)
BrandHeritage	-0.08*** (0.01)			0.02 (0.02)		
VisualAtypicality × BrandHeritage	-0.09*** (0.01)			-0.11*** (0.01)		
<i>Brand-Level Controls</i>						
BrandLuxury	0.65*** (0.01)	0.80*** (0.02)	0.70*** (0.03)	0.42*** (0.02)	0.39*** (0.03)	0.61*** (0.05)
EuropeanBrand	-0.33*** (0.02)	-0.61*** (0.03)	0.18*** (0.05)	-0.16*** (0.03)	-0.61*** (0.05)	0.12 (0.09)
BrandFemininity	-0.26*** (0.01)	-0.33*** (0.01)	0.03 (0.03)	-0.33*** (0.02)	-0.51*** (0.02)	-0.19** (0.06)
BrandSportiveness	-0.03** (0.01)	-0.41*** (0.02)	0.31*** (0.02)	0.04** (0.02)	-0.36*** (0.02)	0.38*** (0.04)
BrandPopularity	1.46*** (0.01)	1.43*** (0.02)	1.39*** (0.03)	1.55*** (0.01)	1.60*** (0.02)	1.51*** (0.05)
<i>Image-Level Controls</i>						
IntraBrandAtypicality	0.04*** (0.01)	0.03* (0.01)	0.08*** (0.02)	-0.04** (0.02)	-0.07*** (0.02)	0.22*** (0.03)
ImageAesthetics	0.01 (0.01)	0.12*** (0.01)	0.02 (0.02)	-0.09*** (0.01)	-0.05** (0.02)	0.00 (0.03)
ImageQuality	0.03*** (0.01)	-0.13*** (0.02)	0.10*** (0.02)	0.15*** (0.01)	0.08*** (0.02)	0.27*** (0.03)
ImageAspectRatio	0.16*** (0.01)	0.04** (0.02)	0.03 (0.02)	0.00 (0.01)	-0.02 (0.02)	-0.06* (0.03)
NumberOfFaces	0.16*** (0.01)	0.34*** (0.02)	0.07*** (0.02)	0.11*** (0.01)	0.17*** (0.02)	0.17*** (0.03)
Sensitive	0.12*** (0.03)	-0.25*** (0.04)	0.39*** (0.06)	0.16*** (0.04)	-0.21*** (0.05)	0.21* (0.10)
<i>Product-Level Controls</i>						
NumberOfProducts	-0.01 (0.01)	-0.05* (0.02)	-0.14*** (0.03)	0.02 (0.02)	0.04 (0.03)	-0.33*** (0.04)
ProductSaturation	0.05*** (0.01)	-0.19*** (0.02)	0.36*** (0.02)	0.08*** (0.02)	-0.11*** (0.02)	0.24*** (0.04)
ProductBrightness	-0.07*** (0.01)	-0.28*** (0.02)	0.16*** (0.02)	0.08*** (0.02)	-0.16*** (0.02)	0.03 (0.04)
ProductContrastOfBrightness	0.09*** (0.01)	-0.09*** (0.01)	0.18*** (0.02)	0.14*** (0.02)	-0.03 (0.02)	0.11*** (0.03)
ProductColorComplexity	-0.05*** (0.01)	0.40*** (0.02)	-0.33*** (0.03)	-0.12*** (0.02)	0.13*** (0.03)	-0.02 (0.05)
ProductWarmth	-0.14*** (0.01)	-0.03* (0.01)	-0.23*** (0.02)	-0.06*** (0.01)	0.02 (0.02)	-0.20*** (0.03)
<i>Text-Level Controls</i>						
WordCount	0.03** (0.01)	0.02 (0.02)	0.09*** (0.02)	0.17*** (0.02)	0.07** (0.02)	0.27*** (0.03)
Hashtags	-0.00 (0.01)	0.07*** (0.01)	-0.06*** (0.02)	0.28*** (0.02)	0.48*** (0.02)	0.08* (0.04)
AtMentions	0.15*** (0.01)	0.04* (0.02)	0.46*** (0.02)	0.15*** (0.02)	0.24*** (0.03)	0.45*** (0.04)
URLs	-0.35*** (0.01)	-0.35*** (0.01)	-0.33*** (0.02)	-0.64*** (0.02)	-0.64*** (0.02)	-0.37*** (0.04)
Emojis	0.07*** (0.01)	-0.01 (0.02)	0.20*** (0.02)	0.12*** (0.02)	0.01 (0.03)	0.27*** (0.05)
TextEmotionality	0.02 (0.01)	0.27*** (0.02)	-0.09*** (0.02)	0.00 (0.01)	0.16*** (0.02)	0.07* (0.03)
<i>Temporal Control</i>						
Weekend	-0.47***	-0.22***	-0.58***	-0.40***	-0.18***	-0.57***

(continued on next page)

Table 3 (continued)

	Main analysis (all tweets)			Robustness check (single-image tweets)		
	Model 1 (All Brands)	Model 2 (Low Heritage Brands)	Model 3 (High Heritage Brands)	Model 1 (All Brands)	Model 2 (Low Heritage Brands)	Model 3 (High Heritage Brands)
	(0.02)	(0.03)	(0.04)	(0.03)	(0.05)	(0.07)
<i>Fixed Effects</i>						
Season Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Theme Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Format Fixed Effects	Yes	Yes	Yes	No	No	No
Constant	2.39*** (0.05)	2.49*** (0.07)	2.13*** (0.09)	2.26*** (0.07)	2.19*** (0.11)	1.83*** (0.15)
Observations	60,669	24,634	10,952	23,418	9,706	4,142

Notes: Standard errors in parentheses; *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

contingent on a brand's heritage level. Specifically, for low-heritage brands, visual atypicality tends to enhance image virality (supporting H_{1a}), whereas, for high-heritage brands, visual atypicality tends to diminish image virality (supporting H_{1b}).

6. Study 2: additional evidence from the automobile industry

To assess the generalizability of our findings beyond the apparel industry, we conducted a replication study using the automobile industry as a focal sector. The methodology closely mirrored that of Study 1, allowing for a direct comparison of results.

6.1. Data

Using Brand Finance's 2022 ranking of the top 500 most valuable brands (Brand Finance, 2022), we identified 28 well-known automobile brands, spanning low-heritage brands (e.g., Tesla, founded in 2003) to high-heritage brands (e.g., Mitsubishi, founded in 1870). These brands were selected due to their market influence and representation of the industry's heritage spectrum. Among these brands, 27 had official international Twitter accounts. In February 2023, we scraped their most recent 3,200 tweets (excluding replies, retweets), retaining only image-featured tweets posted in 2022.

Unlike apparel brands, which feature a diverse range of product categories, automobile brands primarily focus on cars, resulting in a more uniform dataset. We specifically targeted images containing the object "Car" (annotated by the Google Vision API), filtering out irrelevant visuals. This process resulted in 4,087 images presenting cars. Table W9 in Web Appendix H provides an overview of the dataset, including brand names, heritage levels, and the number of sample images for analysis.

6.2. Methods

Most variables were measured similarly to Study 1, with minor industry-specific adjustments for the automobile sector. Since all images feature cars (i.e., only one image theme), we assessed *Visual Atypicality* across the entire dataset, rather than segmenting by theme as was necessary for apparel brands. Examples of typical versus atypical automobile brand images are presented in Table W10 in Web Appendix H. As in Study 1, we conducted a validation analysis to assess the temporal stability of the measure of visual atypicality (see details in Web Appendix H). The results showed that visual atypicality scores remained highly stable across quarters in 2022, confirming consistent industry-level visual norms over the study period.

Luxury Status for automobile brands was assessed using the median price of SUVs as a consistent benchmark, with pricing data sourced primarily from Cars.com. For brands not available in the U.S. market (e.g., Renault), we obtained price information from regional car retail websites (e.g., UK-based sources). We also recorded the country of origin for each car brand. To capture regional industry dominance, we replaced the *European Brand* dummy variable with a *German Brand* dummy variable, acknowledging Germany's strong reputation for high-quality automotive engineering. We excluded brand femininity and brand sportiveness, as these attributes are less relevant in the automobile industry compared to apparel brands. Furthermore, since all images exclusively feature cars as the primary objects, there was no need to control for theme fixed effects in the regression models.

For *Brand Heritage*, we continued using brand age as a proxy, as in Study 1. To determine a perceptual threshold for distinguishing high- vs. low-heritage brands, we conducted a survey similar to that in Study 1, with modifications specific to the automobile industry. Participants ($N = 200$; 43 % female, 56 % male, and 1 % others; $M_{\text{age}} = 31.61$, $SD = 11.34$) were asked to imagine a car brand founded in a given year, ranging from 1870 to 2020, which reflects the actual establishment years in our dataset (1870 to 2013). They then responded to the following prompt: "Imagine a car brand founded in [YEAR, e.g., 1870, 1880, 1890 ... 2020]. It has been operating continuously in the market. Based on this information alone, how much heritage do you think this car brand has? (1 = Low Heritage, 7 = High Heritage)."

As in Study 1, t -tests were conducted for each year to assess whether the perceived heritage value was significantly higher or lower than the midpoint of 4. The results (visualized in Fig. W3 in Web Appendix H) identified 1940 and 1980 as the effective thresholds for distinguishing high-heritage and low-heritage brands, respectively.

Descriptive statistics (mean, SD, minimum, median, and maximum) for all relevant variables across all brands, as well as for low- and high-heritage brands, are provided in Table W12 in [Web Appendix H](#). The pairwise correlations for all continuous variables are shown in Table W13 in [Web Appendix H](#). Following Study 1, we used a negative binomial regression to analyze the automobile brand dataset, first for all brands (*Model 1*) and then for high- and low-heritage subgroups (*Models 2 and 3*). To ensure comparability, all continuous variables (except sharing volume) were standardized as z-scores ($M = 0$, $SD = 1$).

6.3. Results

The estimation results for all three models are displayed in Table W14 in [Web Appendix I](#). First, by examining the full dataset of automobile brands, we found the interaction between visual atypicality and brand heritage is significantly negative ($\beta_{\text{VisualAtypicality} \times \text{BrandHeritage}} = -0.13$, $SE = 0.02$, $p < 0.001$). This pattern suggests that the impact of visual atypicality on image virality depends on a brand's heritage level.

Furthermore, for low-heritage brands, visual atypicality was positively associated with sharing volume ($\beta_{\text{VisualAtypicality}} = 0.35$, $SE = 0.10$, $p < 0.001$), which indicates that consumers are more likely to share atypical visuals from brands with little established heritage, supporting **H_{1a}**. In contrast, for high-heritage brands, the relationship reversed, such that greater visual atypicality corresponded to lower virality ($\beta_{\text{VisualAtypicality}} = -0.20$, $SE = 0.06$, $p < 0.001$). This result indicates that consumers are less likely to share atypical imagery from brands with strong heritage, supporting **H_{2a}**.

6.4. Robustness check

To ensure the robustness of these findings, we conducted additional analyses, as in Study 1, focusing exclusively on single-image tweets to eliminate potential confounds from multi-image posts. The results (see details in Table W14 in [Web Appendix I](#)) mirrored those of the main analysis, confirming that the impact of visual atypicality on image virality depends on brand heritage ($\beta_{\text{VisualAtypicality} \times \text{BrandHeritage}} = -0.11$, $SE = 0.03$, $p < 0.001$). For low-heritage brands, increased visual atypicality continued to show a likelihood of enhancing image virality ($\beta_{\text{VisualAtypicality}} = 0.33$, $SE = 0.14$, $p = 0.02$), whereas, for high-heritage brands, increasing atypicality tended to reduce virality ($\beta_{\text{VisualAtypicality}} = -0.46$, $SE = 0.10$, $p < 0.001$).

6.5. Discussion

This replication study provides additional empirical support for the moderating effect of brand heritage on the relationship between visual atypicality and image virality in the automobile industry. The results mirror those observed in Study 1. For low-heritage

Table 4
Experiment materials.

	Typical	Atypical
Female version	 (A)	 (B)
Male version	 (C)	 (D)

brands, visual atypicality tends to enhance image virality (supporting H_{1a}); however, for high-heritage brands, visual atypicality tends to reduce image virality (supporting H_{2a}). These findings increase the generalizability of the observed effect across brand categories.

7. Study 3: examining the underlying mechanisms

To replicate the effects observed in the field, establish causal relationships, and examine the underlying psychological mechanisms, we conducted Study 3, an online experiment using a fictitious apparel brand. Apparel brands were chosen for this study because they sell widely used personal goods (e.g., clothing, shoes, accessories), making their content highly relevant to all participants. This familiarity should also make it easier for participants to judge whether brand imagery is visually typical or atypical.

7.1. Preparation of typical vs. atypical images

Drawing on insights from Study 1, we observed that typical apparel brand images usually feature a model outfitted in the brand's topwear, bottomwear, and footwear, positioned centrally against a plain background. To mirror these characteristics, we selected two images, one with a female model and the other with a male model, each centered against a neutral gray background (see Posts A and C in Table 4). For the atypical condition, we manipulated the models' postures to create unconventional visuals. In the atypical condition, the female model is depicted in a reclined pose, balancing on her hands with one leg raised high to evoke an unexpected sense of movement (see Post B in Table 4); and the male model is shown mid-leap, with one leg bent and his arms extended in a dynamic, expressive stance (see Post D in Table 4).

A pretest with participants from Prolific ($N = 100$, 50 % female, 50 % male; $M_{age} = 33.99$, $SD = 11.31$), who were randomly assigned to view one of four images, confirmed that our manipulation of visual atypicality effectively made atypical images appear more atypical (In general, the visual style of this tweet's image is atypical for an apparel brand, i.e., not commonly seen in images shared by other apparel brands; [1 = strongly disagree and 7 = strongly agree]; $M_{typical} = 3.63$ vs. $M_{atypical} = 4.48$; $t(98) = -2.15$, $p = 0.03$). More importantly, there were no significant differences in image quality (The image is of high quality; [1 = strongly disagree and 7 = strongly agree]; $M_{typical} = 4.92$ vs. $M_{atypical} = 4.94$; $t(98) = -0.08$, $p = 0.94$), overall aesthetics (The image is visually pleasing; [1 = strongly disagree and 7 = strongly agree]; $M_{typical} = 5.08$ vs. $M_{atypical} = 4.65$; $t(98) = 1.47$, $p = 0.14$), or informativeness (The image is informative; [1 = strongly disagree and 7 = strongly agree]; $M_{typical} = 3.50$ vs. $M_{atypical} = 3.37$; $t(98) = 0.42$, $p = 0.67$).

7.2. Participants, design, and procedure

Eight hundred and eighty regular Twitter users were recruited from Prolific for a pre-registered study (<https://aspredicted.org/zpzb-7t97.pdf>), which employed a 2 (image type: typical vs. atypical) $\times$ 2 (brand heritage: low vs. high) between-subjects design. Nine failed an attention check requiring at least a three-word image content description, leaving 871 participants (53.27 % female, 44.78 % male, and 1.95 % others; $M_{age} = 32.54$, $SD = 10.80$; OSF repository: <https://osf.io/m8235/>).

Participants first read a brief brand introduction designed to manipulate brand heritage. To avoid biases linked to established brands, we introduced the fictitious brand "Lumiens." In the low-heritage condition, Lumiens was described as being founded in 2023 and positioned as an emerging apparel brand. In the high-heritage condition, it was described as having been established in 1883 and positioned as a heritage brand. To enhance ecological validity, participants were instructed to imagine that they followed Lumiens on Twitter and enjoyed recommending the brand to others.

Subsequently, participants viewed a tweet from Lumiens featuring either a typical or atypical image, as described above. To ensure the image was relevant, the social media post was tailored to participants' self-disclosed gender. After indicating their gender (1 = male, 2 = female, and 3 = non-binary), male participants saw the male version of the post, female participants saw the female version, and non-binary participants were randomly assigned to one of the two versions.

After viewing the tweet, participants completed a three-item scale ($\alpha = 0.97$) adapted from Nie et al. (2022) to assess their willingness to share the tweet: (1) *How likely are you to share this tweet with others?* (1 = very unlikely and 7 = very likely), (2) *How seriously would you consider sharing this tweet with others?* (1 = very unserious and 7 = very serious), and (3) *How interested are you in sharing this tweet with others?* (1 = very uninterested and 7 = very interested). The two proposed mediators, perceived entertainment and expectation alignment, were measured using validated scales adapted from prior research. Perceived entertainment was assessed with two items ($r = 0.81$) adapted from Schweidel et al. (2006): (1) *The image in this tweet is entertaining*, and (2) *The image in this tweet is interesting* (1 = not at all and 7 = very much). Expectation alignment was measured using five items ($\alpha = 0.95$) from Montaguti et al. (2023): (1) *This tweet image is coherent with the typical contents of the brand*, (2) *This tweet image is in line with the personality and values of the brand*, (3) *This tweet image is coherent with the brand*, (4) *This tweet image effectively represents the brand*, and (5) *In general, this tweet image is consistent with what I would expect from the brand* (1 = strongly disagree and 7 = strongly agree).

To assess the effectiveness of our manipulations, we measured two additional variables. First, perceived visual atypicality was assessed using an item adapted from Schnurr (2017): *In general, the visual style of this tweet's image is atypical for an apparel brand* (i.e., not commonly seen in images shared by other apparel brands (1 = strongly disagree and 7 = strongly agree)). Second, perceived brand heritage was measured with the statement: *Lumiens is a brand with a long-standing history* (1 = strongly disagree and 7 = strongly agree).

7.3. Manipulation checks

A 2 (visual atypicality) $\times$ 2 (brand heritage) ANOVA on perceived image atypicality revealed a significant main effect of visual

atypicality ($F(1, 867) = 30.15, p < 0.001$), but no significant effect of brand heritage ($F(1, 867) = 2.39, p = 0.12$) or its interaction with visual atypicality ($F(1, 867) = 2.86, p = 0.09$). Participants in the atypical condition ($M = 4.49$) rated the image as more atypical than those in the typical condition ($M = 3.85$), confirming the effectiveness of the visual atypicality manipulation.

Another 2 (visual atypicality) $\times$ 2 (brand heritage) ANOVA on perceived brand heritage showed a significant main effect of brand heritage ($F(1, 867) = 130.01, p < 0.001$), whereas neither visual atypicality ($F(1, 867) = 0.00, p = 0.98$) nor its interaction with brand heritage had a significant effect ($F(1, 867) = 0.12, p = 0.73$). Participants in the low heritage condition ($M = 4.14$) perceived the brand as having lower heritage than those in the high heritage condition ($M = 5.33$), indicating that the manipulation of brand heritage was successful.

7.4. Main effects on willingness to share

We conducted a 2 (visual atypicality) $\times$ 2 (brand heritage) ANOVA on willingness to share, which revealed no significant main effects of either visual atypicality ($F(1, 867) = 0.06, p = 0.81$) or brand heritage ($F(1, 867) = 2.81, p = 0.09$). However, there was a significant visual atypicality $\times$ brand heritage interaction ($F(1, 867) = 10.86, p < 0.001$), indicating that the effect of visual atypicality on willingness to share depended on brand heritage.

Separate analyses confirmed this pattern: For the low-heritage brand, participants had a greater willingness to share when exposed to visually atypical imagery ($M = 4.23$) compared to typical imagery ($M = 3.72$; $F(1, 433) = 6.02, p = 0.01$). These results support **H_{1a}**. In contrast, for the high-heritage brand, willingness to share was lower in the atypical condition ($M = 3.51$) than in the typical condition ($M = 3.95$; $F(1, 434) = 4.86, p = 0.03$), supporting **H_{2a}**.

7.5. Mediation analysis

To confirm that the independent variable (visual atypicality) and the mediators (perceived entertainment and expectation alignment) are conceptually distinct, we examined their correlations. The results showed low correlations ($r = 0.17$ for visual atypicality and entertainment; $r = -0.10$ for visual atypicality and expectation alignment), suggesting that the constructs are sufficiently distinct.

Mediation for low heritage brands. For low-heritage brands, a one-way ANOVA showed that participants exposed to visually atypical images ($M = 4.77$) reported significantly higher perceived entertainment than those viewing typical images ($M = 3.99$; $F(1, 433) = 21.01, p < 0.001$). However, there was no significant difference in expectation alignment between the visually atypical ($M = 5.04$) and typical ($M = 5.12$) conditions ($F(1, 433) = 0.42, p = 0.52$).

Then, we conducted a mediation analysis using PROCESS Model 4 with 10,000 bootstrap samples (Hayes, 2017), specifying visual atypicality as the independent variable, willingness to share as the dependent variable, and both perceived entertainment and expectation alignment as parallel mediators. The results revealed a significant indirect effect through perceived entertainment ($b = 0.59, SE = 0.14$; $CI_{95\%}: [0.33, 0.86]$), while the indirect effect through expectation alignment was not significant ($b = -0.03, SE = 0.04$; $CI_{95\%}: [-0.12, 0.06]$).

Given that expectation alignment did not differ significantly between conditions, we additionally conducted a mediation analysis focusing solely on perceived entertainment as the mediator (PROCESS Model 4 with 10,000 bootstrap samples). The results showed a significant indirect effect of visual atypicality on willingness to share through perceived entertainment ($b = 0.72, SE = 0.16$; $CI_{95\%}: [0.40, 1.04]$). These results support **H_{1b}**.

Mediation for high heritage brands. For the high-heritage brand, a one-way ANOVA showed that participants exposed to visually atypical images ($M = 4.42$) reported significantly higher perceived entertainment than those viewing typical images ($M = 3.99$; $F(1, 434) = 6.48, p = 0.01$). Besides, they perceived atypical images ($M = 4.72$) as significantly less aligned with the brand's expected communication style compared to typical images ($M = 5.15$; $F(1, 434) = 11.85, p < 0.001$).

To test the pathways, we first used PROCESS Model 4 with 10,000 bootstrap samples, specifying visual atypicality as the independent variable, willingness to share as the dependent variable, and perceived entertainment and expectation alignment as parallel mediators. The results showed opposing indirect effects: atypical visuals increased perceived entertainment, boosting willingness to share ($b = 0.31, SE = 0.13$; $CI_{95\%}: [0.07, 0.56]$), but also reduced expectation alignment, which in turn lowered willingness to share ($b = -0.11, SE = 0.05$; $CI_{95\%}: [-0.22, -0.03]$). Notably, the total effect of visual atypicality on willingness to share remained significantly negative after accounting for these opposing mediators ($b = -0.44, p = 0.03$). These results suggest that while atypical visuals may enhance entertainment, this benefit is insufficient to offset the harm caused by violating expectations of how a heritage brand should communicate, ultimately reducing consumers' willingness to share.

To further isolate this pathway, we ran an additional mediation analysis using PROCESS Model 4 with expectation alignment as the sole mediator. This analysis confirmed a substantial negative indirect effect through expectation alignment ($b = -0.37, SE = 0.11$; $CI_{95\%}: [-0.60, -0.16]$). These results support **H_{2b}**.

7.6. Discussion

The experimental findings provide robust causal evidence for the effects observed in the field studies (supporting **H_{1a}** and **H_{2a}**). More importantly, they validate the proposed psychological mechanisms: heightened visual atypicality increases the virality of low-heritage brands' images by enhancing perceived entertainment (supporting **H_{1b}**), but reduces the virality of high-heritage brands' images by disrupting consumer expectations (supporting **H_{2b}**).

8. General discussion

Three studies provide converging evidence that the effect of visual atypicality on image virality depends on brand heritage: for low-heritage brands, atypical visuals increase sharing by enhancing perceived entertainment, whereas for high-heritage brands, they reduce sharing due to expectation misalignment.

8.1. Theoretical contributions

This research makes four important contributions to the literature on content marketing and social media marketing strategy. First, inspired by prior marketing research on the atypicality of product design (e.g., Landwehr et al., 2011), this research introduces the concept of *visual atypicality*, an increasingly relevant yet seldom-researched construct, and differentiates it from related concepts such as *visual novelty* and *visual creativity*. As brands increasingly rely on visual content to communicate with consumers (e.g., ranging from images on Instagram to videos on TikTok, and potentially even virtual-reality imagery in the metaverse; Bao et al., 2024; Li et al., 2019; Ordenes et al., 2019), it becomes crucial for marketers to examine how certain visual characteristics influence consumer perception and engagement. Despite its potential significance in brands' visual communication practice, *visual atypicality* and its implications for marketing practice have received little attention in prior research. By introducing and conceptualizing *visual atypicality*, this work lays the groundwork for future researchers to explore its role in shaping consumer responses, brand perceptions, and engagement outcomes.

Second, this work advances the understanding that a brand's content effectiveness is shaped not only by its intrinsic features but also by its relationship to industry norms. Much of the existing literature has focused on isolated content features. For instance, in textual content, studies have examined emotional tone (Berger & Milkman, 2012), emoji usage in text (McShane et al., 2021), and linguistic certainty (Pezzuti et al., 2021). More closely aligned with our research's focus on image content, prior studies have explored visual aspects such as color complexity (Kanuri et al., 2024) and visual complexity (Overgoor et al., 2022). However, these studies have not fully accounted for the broader industry context in which brands operate. Our research fills this gap by demonstrating that consumer engagement is also significantly impacted by the extent to which a brand's content aligns with or diverges from the prevailing visual practices within its industry. This insight encourages future research to go beyond analyzing individual content features in isolation and instead examine how a brand's content interacts with and is shaped by the broader competitive landscape.

Third, this research challenges the widespread assumption that content strategies exert a uniform influence across all brands (e.g., Cruz et al., 2017; McShane et al., 2021; Pancer et al., 2019; Pezzuti et al., 2021). While some studies have explored moderating factors (e.g., Pancer et al., 2019; Pezzuti et al., 2021), these investigations typically focus on how certain conditions amplify or weaken a given effect rather than on cases where a strategy may completely backfire. Rather than endorsing a one-size-fits-all approach, our work identifies brand heritage as a critical boundary condition that determines the success of visual atypicality. Our findings suggest that unconventional visuals can enhance engagement for low-heritage brands, while in stark contrast, the same approach may undermine engagement for high-heritage brands. This understanding highlights the importance of tailoring marketing strategies to align with a brand's identity (e.g., heritage), urging future researchers to explore situations in which a seemingly effective strategy may, in fact, backfire.

Finally, this research extends the literature on consumer digital word-of-mouth and sharing behavior by identifying two key psychological drivers: the entertainment value of content and its alignment with brand-related expectations. While breaking norms to appear atypical is often praised for helping brands stand out in a crowded digital marketplace, our findings reveal the double-edged nature of this strategy. On one hand, unconventional visuals can captivate audiences and increase sharing by enhancing entertainment value, consistent with prior research showing that creative ads boost engagement by increasing perceived intrigue (Smith et al., 2008; X. Yang & Smith, 2009). On the other hand, excessive deviation from a brand's established norms can alienate consumers, echoing findings that atypical product designs reduce consumer liking (Landwehr et al., 2011) and atypical image pairings with review text lower perceived review helpfulness (Ceylan et al., 2023). In sum, this research contributes to ongoing debates on the boundary conditions of creativity in consumer behavior (Rosengren et al., 2020), and highlights the need for a balanced approach that combines innovation with consistency.

8.2. Methodological contributions

Although *visual atypicality* shares some conceptual similarities with *design atypicality*, which has been widely studied in marketing research (Desai & Ratneshwar, 2003; Landwehr et al., 2011, 2013; Schnurr, 2017), it remains a relatively new concept. Consequently, there is no established, widely documented methodology available to measure it, particularly in a way that allows for large-scale, automated assessment. Drawing from prior research on design atypicality, we developed a machine-learning-based measurement method that captures visual atypicality in brand-generated images. This approach effectively controls for content consistency by ensuring that all images being compared depict the same key content. A validation study confirmed the method's high accuracy in capturing visual atypicality as perceived by human judgment.

With the vast availability of brand-generated content in digital spaces, this machine-learning-based methodology offers a scalable solution for future research. It can be adopted to explore how visual atypicality in brand-generated images or other forms of visual content (e.g., videos) influences consumer reactions, such as content engagement, product evaluation, purchase intention, and brand perception.

8.3. Managerial implications

Apart from the theoretical and methodological contributions, the findings also bring three actionable managerial implications that marketing practitioners can implement to optimize social media engagement. First, for brands with a relatively short history or emerging market presence, our research demonstrates that adopting visually atypical content can significantly boost consumer sharing by enhancing the entertainment value of the content. Managers could encourage their creative teams to experiment with unconventional visual elements that break away from industry norms, such as unique camera angles, dynamic backgrounds, or unexpected model poses. By doing so, low-heritage brands can differentiate themselves in a crowded digital marketplace and potentially increase image content virality.

In contrast, high-heritage brands, with their established legacy and consistent consumer expectations, are more vulnerable to the negative effects of radical visual changes. Our findings suggest that when such brands stray too far from the traditional and typical visual presentation approach, it can lead to misalignment with consumer expectations and ultimately reduce consumers' willingness to share. Therefore, high-heritage brands should focus on maintaining visual consistency with the industry norms to preserve authenticity while introducing innovation in a measured and incremental manner. Pilot campaigns, limited-time visual experiments, or subtle modern updates can help these brands evolve without alienating their loyal customer base.

Regardless of brand heritage, it is crucial for marketing practitioners to ground their visual strategies in robust consumer research. Our research underscores the value of understanding how visual elements affect engagement differently across brand types. By investing in regular consumer surveys, focus groups, and rigorous A/B testing, brands can gain actionable insights into which visual elements resonate with their audience. This iterative approach enables brands to continuously refine their content, ensuring that any visual atypicality introduced either enhances entertainment value for low-heritage brands or maintains alignment with expectations for high-heritage brands.

8.4. Limitations and future research directions

Despite the valuable insights provided by this research, several limitations must be acknowledged, each of which presents promising opportunities for future inquiry. First, although our research employed advanced machine-learning techniques to quantify visual atypicality, these computational measures may not fully encapsulate the complexity of human aesthetic perception. Our validation experiment indicated acceptable accuracy; however, the measurement could be further refined as technology evolves, for example, by leveraging more sophisticated image analysis methods or combining computational assessments with richer human judgment data.

Second, visual atypicality in our research was conceptualized and measured at the industry level, that is, as a deviation from the visual norms established collectively by all brands within a given industry. However, we acknowledge that within each industry, brands can be further categorized into meaningful subgroups (e.g., luxury vs. premium vs. mainstream; masculine vs. feminine; outdoorsy vs. indoorsy), each potentially operating under its own set of visual norms. As a result, what appears atypical at the industry level may be perceived differently within a more narrowly defined subgroup. Future research could develop subgroup-specific benchmarks of visual typicality and investigate how deviations from those subgroup norms influence consumer reactions. Such an approach would offer a more granular understanding of how visual atypicality functions within diverse brand positioning strategies and consumer expectations.

Third, it is notable that although visual norms within a given industry may remain relatively stable over a certain period, particularly in the absence of major technological, cultural, or social shifts, these norms can be mutable. Over longer time spans or during periods of rapid change (e.g., evolving consumer aesthetics or shifts in cultural values), industry-level visual expectations may evolve. Future research could adopt a longitudinal approach to investigate how visual norms evolve over time and how brands can adapt their visual strategies accordingly to maintain engagement and relevance.

Lastly, the experimental design implemented in Study 3, while instrumental in establishing causality, was limited by its reliance on Twitter users. This sample may not entirely represent the diverse spectrum of social media users, potentially affecting the external validity of our results. Future research would benefit from adopting more diverse sampling methods, including field experiments or longitudinal studies, to observe how the effects of visual atypicality evolve over time and across different demographic segments.

CRediT authorship contribution statement

Zitian Adam: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhiyong Yang:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijresmar.2025.07.005>.

Data availability

Data will be made available on request.

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